

Question	Answer	Marks
4(a)	(field line indicates) direction of force	B1
	force on a positive charge	B1
4(b)(i)	one straight line perpendicular to plates, starting on one plate and finishing on the other	B1
	five straight lines perpendicular to plates between the plates, uniformly spaced	B1
	downwards arrows on lines	B1
4(b)(ii)	$E = V / d$	C1
	$= 2400 / 0.046$	A1
	$= 5.2 \times 10^4 \text{ N C}^{-1}$	
4(c)(i)	smooth curve in region of field and straight line outside field	B1
	direction of deflection shown as downwards in region of field	B1
4(c)(ii)	helium nucleus has double the charge but four times the mass	B1
	velocity parallel to plates same and acceleration perpendicular to plates smaller (for helium)	B1
	final speed is lower (for helium)	B1

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5(a)(i)	0.01	0.1